

MODULE 2

ENERGY EFFICIENCY IN ELECTRICAL SYSTEMS

INTRODUCTION TO ELECTRIC POWER SUPPLY SYSTEMS

Electric power supply system in a country comprises of

- Generating units that produce electricity
- High voltage transmission lines that transport electricity over long distances
- Distribution lines that deliver the electricity to consumers
- Substations that connect the pieces to each other and
- Energy control centers to coordinate the operation of the components.

The Figure 1 shows a simple electric supply system with transmission and distribution network and linkages from electricity sources to end-user.

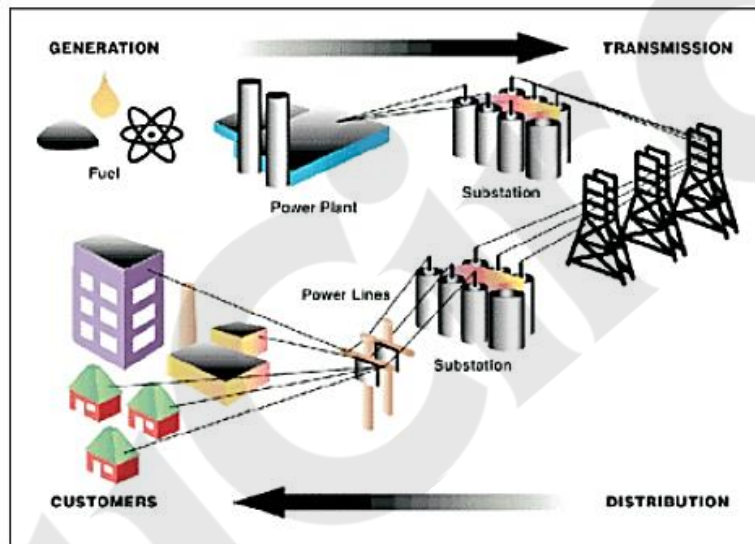


Figure 1: Typical Electric Power Supply Systems

POWER GENERATION PLANT:

- The fossil fuels such as coal, oil and natural gas, nuclear energy, and falling water (hydel) are commonly used energy sources in the power generating plant.
- A wide and growing variety of unconventional generation technologies and fuels have also been developed, including cogeneration, solar energy, wind generators, and waste materials.
- About 70 % of power generating capacity in India is from coal based thermal power plants.
- The principle of coal-fired power generation plant is shown in figure 2.
- Energy stored in the coal is converted in to electricity in thermal power plant.
- Coal is pulverized to the consistency of talcum powder.
- Then powdered coal is blown into the water wall boiler where it is burned at temperature higher than 1300°C.
- The heat in the combustion gas is transferred into steam.

- This high-pressure steam is used to run the steam turbine to spin.
- Finally, turbine rotates the generator to produce electricity.

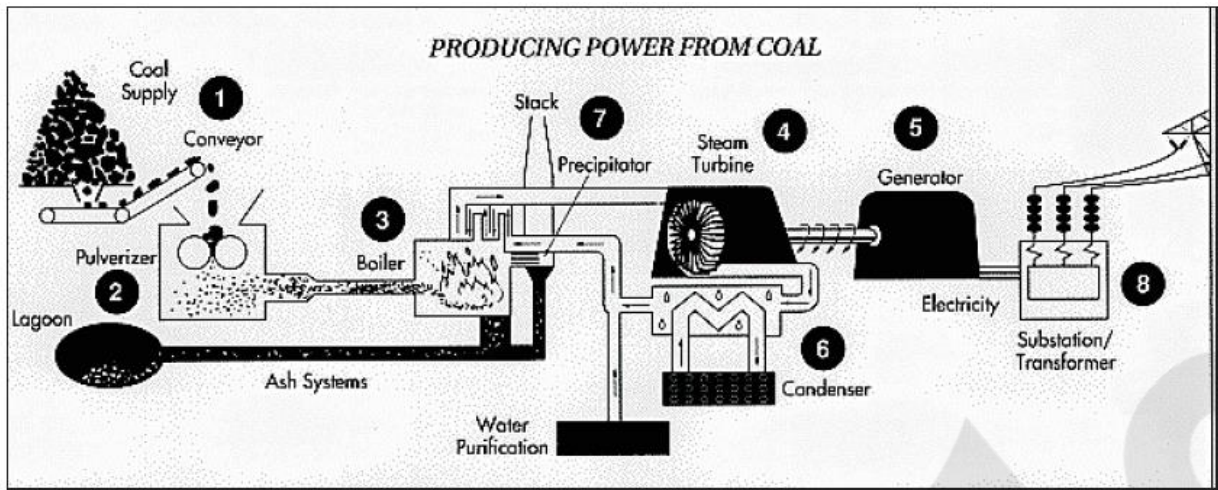


Figure 2: Principle of Thermal Power Generation

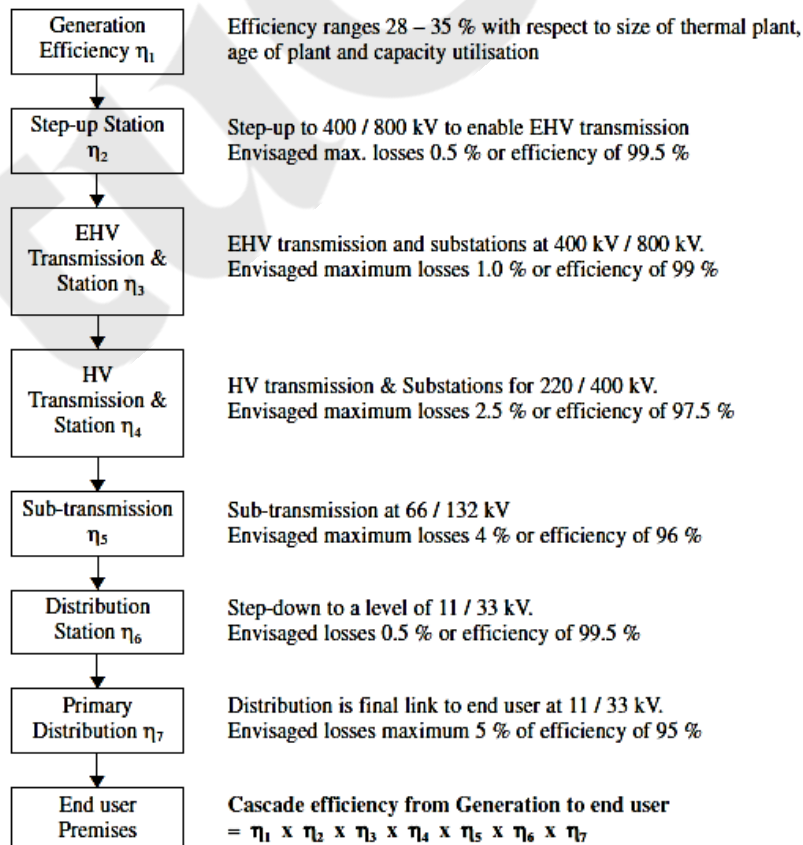
- In India, for the coal based power plants, the overall efficiency ranges from 28% to 35% depending upon the size, operational practices and capacity utilization.

TRANSMISSION AND DISTRIBUTION LINES:

- Power plants generate the AC power at 50 Hz frequency, with the voltages between 11kV and 33kV.
- The Power generated is stepped up to High voltage and Extra high voltage for transmission purpose over a long distance i.e 400 kV or 220 kV minimize the losses (Primary transmission).
- This is further step down 132 kV/110kV/66kV or 33 kV at receiving station (Secondary transmission).
- At substation this voltage is reduced to 11kV/6.6kV/3.3kV, these voltage is supplied for bulk consumers (Primary Distribution).
- Then distribution voltage is again stepped down to 440 and 230 volts.
- This power is then supplied for commercial and residential consumer through distribution lines. (Secondary Distribution).
- There is no difference between a transmission line and a distribution line except for the voltage level and power handling capability.
- Transmission lines are usually capable of transmitting large quantities of electric energy over great distances. They operate at high voltages. Distribution lines carry limited quantities of power over shorter distances.
- The power loss in line is proportional to resistance and square of current. (i.e. $P_{loss} = I^2 R$).
- Higher voltage transmission and distribution thus would help to minimize line voltage drop in the ratio of voltages, and the line power loss in the ratio of square of voltages.
- Lower voltage transmission and distribution also calls for bigger size conductor on account of current handling capacity needed.

CASCADE EFFICIENCY:

- The primary function of transmission and distribution equipment is to transfer power economically and reliably from one location to another.
- Conductors in the form of wires and cables strung on towers and poles carry the high-voltage, AC electric current.
- A large number of copper or aluminum conductors are used to form the transmission path. Energy loss in transmission lines is wasted in the form of I^2R losses.
- Capacitors are used to correct power factor by causing the current to lead the voltage. When the AC currents are kept in phase with the voltage, operating efficiency of the system is maintained at a high level.
- Circuit-interrupting devices are switches, relays, circuit breakers, and fuses. Each of these devices is designed to carry and interrupt certain levels of current. Making and breaking the current carrying conductors in the transmission path with a minimum of arcing is one of the most important characteristics of this device. Relays sense abnormal voltages, currents, and frequency and operate to protect the system.
- Transformers are placed at strategic locations throughout the system to minimize power losses in the T&D system. They are used to change the voltage level from low-to-high in step-up transformers and from high-to-low in step-down units.
- If we consider the electricity flow from generation to the user in terms of cascade energy efficiency, typical cascade efficiency profile from generation to 11 – 33 kV user will be as below:



- The cascade efficiency in the T&D system from output of the power plant to the end use is 87% (i.e. $0.995 \times 0.99 \times 0.975 \times 0.96 \times 0.995 \times 0.95 = 87\%$).

INDUSTRIAL END USER:

- At the industrial end user premises, again the plant network elements like transformers at receiving sub-station, switchgear, lines and cables, load-break switches, capacitors cause losses, which affect the input-received energy. However the losses in such systems are meager and unavoidable.
- ONE Unit saved = TWO Units Generated
- After power generation at the plant it is transmitted and distributed over a wide network.
- The standard technical losses are around 17 % in India (Efficiency = 83%). But the figures for many of the states show T & D losses ranging from 17 – 50 %.
- All these may not constitute technical losses, since un-metered and pilferage are also accounted in this loss.
- When the power reaches the industry, it meets the transformer. The energy efficiency of the transformer is generally very high.
- Next, it goes to the motor through internal plant distribution network. A typical distribution network efficiency including transformer is 95% and motor efficiency is about 90%.
- Another 30 % (Efficiency = 70%) is lost in the mechanical system which includes coupling/ drive train, a driven equipment such as pump and flow control valves/throttling etc.
- Thus the overall energy efficiency becomes 50%. ($0.83 \times 0.95 \times 0.9 \times 0.70 = 0.50$, i.e. 50% efficiency)
- Hence one unit saved in the end user is equivalent to two units generated in the power plant. ($1 \text{ Unit} / 0.5 \text{ Eff} = 2 \text{ Units}$)

ELECTRICITY BILLING:

- The electricity billing by utilities for medium & large enterprises, in High Tension (HT) category, is often done on two-part tariff structure, i.e. one part for capacity or demand drawn and the second part for actual energy drawn during the billing cycle.
- Capacity or demand is in kVA (apparent power) or kW terms.
- The reactive energy (i.e.) kVA_{rh} drawn by the service is also recorded and billed for in some utilities, because this would affect the load on the utility.
- Accordingly, utility charges for maximum demand, active energy and reactive power drawn (as reflected by the power factor) in its billing structure. In addition, other fixed and variable expenses are also levied.
- **The tariff structure generally includes the following components:**
 - a) Maximum demand Charges: These charges relate to maximum demand registered during month/billing period and corresponding rate of utility.

- b) Energy Charges: These charges relate to energy (kilowatt hours) consumed during month / billing period and corresponding rates, often levied in slabs of use rates. Some utilities now charge on the basis of apparent energy (kVAh), which is a vector sum of kWh and kVArh.
- c) Power factor penalty or bonus rates, as levied by most utilities, are to contain reactive power drawn from grid.
- d) Fuel cost adjustment charges as levied by some utilities are to adjust the increasing fuel expenses over a base reference value.
- e) Electricity duty charges levied w.r.t units consumed.
- f) Meter rentals
- g) Lighting and fan power consumption is often at higher rates, levied sometimes on slab basis or on actual metering basis.
- h) Time of Day (TOD) rates like peak and non-peak hours are also prevalent in tariff structure provisions of some utilities.
- i) Penalty for exceeding contract demand.
- j) Surcharge if metering is at LT side in some of the utilities.
- Analysis of utility bill data and monitoring its trends helps energy manager to identify ways for electricity bill reduction through available provisions in tariff framework, apart from energy budgeting.
- The utility employs an electromagnetic or electronic trivector meter, for billing purposes.
- **The minimum outputs from the electromagnetic meters are:**
 - Maximum demand registered during the month, which is measured in preset time intervals (say of 30-minute duration) and this is reset at the end of every billing cycle.
 - Active energy in kWh during billing cycle
 - Reactive energy in kVArh during billing cycle and
 - Apparent energy in kVAh during billing cycle
- It is important to note that while maximum demand is recorded, it is not the instantaneous demand drawn, but the time integrated demand over the predefined recording cycle.

Example:

In an industry, if the drawl over a recording cycle of 30 minutes is:

2500 kVA for 4 minutes

3600 kVA for 12 minutes

4100 kVA for 6 minutes

3800 kVA for 8 minutes

Solution:

The MD recorder will be computing MD as

$$\frac{(2500 * 4) + (3600 * 12) + (4100 * 6) + (3800 * 8)}{30} = 3606.67\text{kVA}$$

- The month's maximum demand will be the highest among such demand values recorded over the month.
- The meter registers only if the value exceeds the previous maximum demand value and thus, even if, average maximum demand is low, the industry / facility has to pay for the maximum demand charges for the highest value registered during the month, even if it occurs for just one recording cycle duration i.e., 30 minutes during whole of the month.
- **Demand Curve:**

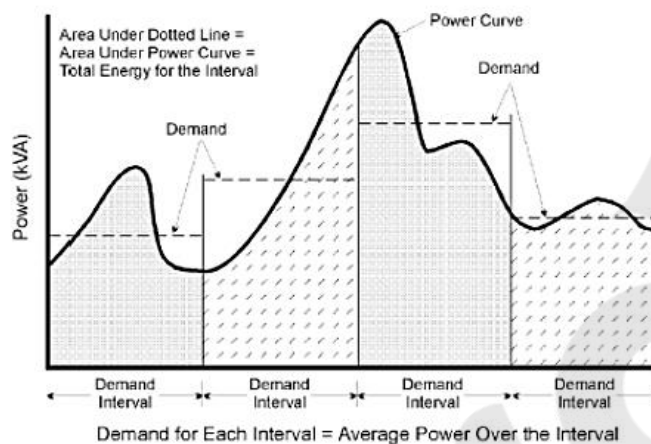


Figure: Demand Curve

- A typical demand curve is shown in Figure. It is observed that the demand varies from time to time. The demand is measured over predetermined time interval and averaged out for that interval as shown by the horizontal dotted line.
- Most electricity boards have changed over from conventional electromechanical trivector meters to electronic meters, which have some excellent provisions that can help the utility as well as the industry. These provisions include:
 - Substantial memory for logging and recording all relevant events
 - High accuracy up to 0.2 class
 - Amenability to time of day tariffs
 - Tamper detection /recording
 - Measurement of harmonics and Total Harmonic Distortion (THD)
 - Long service life due to absence of moving parts
 - Amenability for remote data access/downloads
- Trend analysis of purchased electricity and cost components can help the industry to identify key result areas for bill reduction within the utility tariff available framework along the following lines.

ELECTRICAL LOAD MANAGEMENT AND MAXIMUM DEMAND CONTROL

Need for Electrical Load Manage:

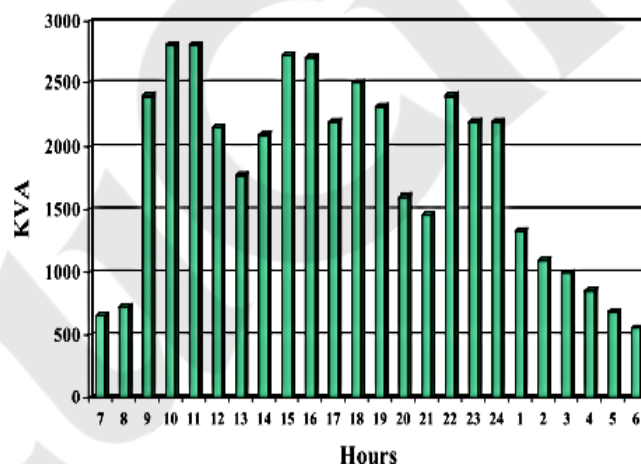
- The growth in the electricity use and diversity of end use segments in time of use has led to shortfalls in capacity to meet demand.

- As capacity addition is costly and only a long time prospect, better load management at user end helps to minimize peak demands on the utility infrastructure as well as better utilization of power plant capacities.
- The utilities (State Electricity Boards) use power tariff structure to influence end user in better load management through measures like time of use tariffs, penalties on exceeding allowed maximum demand, night tariff concessions etc.
- The growth in the electricity use and diversity of end use segments in time of use has led to shortfalls in capacity to meet demand.
- Load management is a powerful means of efficiency improvement both for end user as well as utility.
- As the demand charges constitute a considerable portion of the electricity bill, from user angle too there is a need for integrated load management to effectively control the maximum demand.

Step by Step Approach for Maximum Demand Control

1. Load Curve Generation:

- Presenting the load demand of a consumer against time of the day is known as a Load curve.
- If it is plotted for the 24 hours of a single day, it is known as an Hourly Load curve.
- If daily demands plotted over a month, it is called Daily Load curves.
- A typical hourly load curve for an engineering industry is shown in Figure.



2. Rescheduling of Loads:

- Rescheduling of large electric loads and equipment operations, in different shifts can be planned and implemented to minimize the simultaneous maximum demand.
- For this purpose, it is advisable to prepare an operation flow chart and a process chart.
- Analyzing these charts and with an integrated approach, it would be possible to reschedule the operations and running equipment in such a way as to improve the load factor which in turn reduces the maximum demand.

3. Storage of Products/in process material/process utilities like refrigeration:

- It is possible to reduce the maximum demand by building up storage capacity of products/ materials, water, chilled water / hot water, using electricity during off peak periods.

- Off peak hour operations also help to save energy due to favorable conditions such as lower ambient temperature etc.
- Example: Ice bank system is used in milk & dairy industry. Ice is made in lean period and used in peak load period and thus maximum demand is reduced.

4. Shedding of Non-Essential Loads:

- When the maximum demand tends to reach preset limit, shedding some of non-essential loads temporarily can help to reduce it.
- It is possible to install direct demand monitoring systems, which will switch off non-essential loads when a preset demand is reached.
- Simple systems give an alarm, and the loads are shed manually.
- Sophisticated microprocessor controlled systems are also available, which provide a wide variety of control options like:
 - ✓ Accurate prediction of demand
 - ✓ Graphical display of present load, available load, demand limit
 - ✓ Visual and audible alarm
 - ✓ Automatic load shedding in a predetermined sequence
 - ✓ Automatic restoration of load
 - ✓ Recording and metering

5. Operation of Captive Generation and Diesel Generation Sets:

- When diesel generation sets are used to supplement the power supplied by the electric utilities, it is advisable to connect the D.G. sets for durations when demand reaches the peak value.
- This would reduce the load demand to a considerable extent and minimize the demand charges.

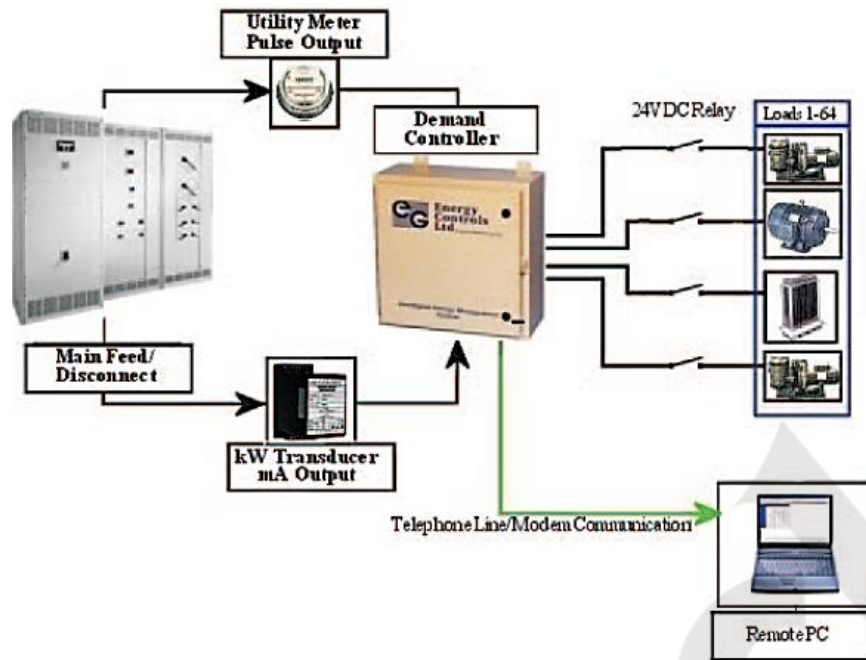
6. Reactive Power Compensation

- The maximum demand can also be reduced at the plant level by using capacitor banks and maintaining the optimum power factor.
- Capacitor banks are available with microprocessor based control systems.
- These systems switch on and off the capacitor banks to maintain the desired Power factor of system and optimize maximum demand thereby

MAXIMUM DEMAND CONTROLLERS:

- High-tension (HT) consumers have to pay a maximum demand charge in addition to the usual charge for the number of units consumed.
- This charge is usually based on the highest amount of power used during some period (say 30 minutes) during the metering month.
- The maximum demand charge often represents a large proportion of the total bill and may be based on only one isolated 30-minute episode of high power use.

- Considerable savings can be realised by monitoring power use and turning off or reducing non-essential loads during such periods of high power use.



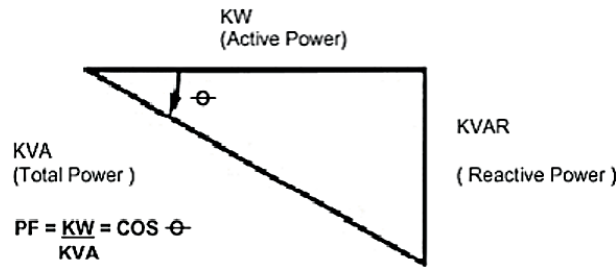
- Maximum Demand Controller is a device designed to meet the need of industries conscious of the value of load management as shown in figure.
- Alarm is sounded when demand approaches a preset value.
- If corrective action is not taken, the controller switches off non-essential loads in a logical sequence.
- This sequence is predetermined by the user and is programmed jointly by the user and the supplier of the device.
- The plant equipments selected for the load management are stopped and restarted as per the desired load profile.
- Demand control scheme is implemented by using suitable control contactors.
- Audio and visual annunciations could also be used.

POWER FACTOR IMPROVEMENT AND BENEFITS:

Power Factor Basics

- In all industrial electrical distribution systems, the major loads are resistive and inductive.
- Resistive loads are incandescent lighting.
- In case of pure resistive loads, the voltage (V), current (I), resistance (R) relations are linearly related, i.e $V = I \times R$ and $P = V \times I$
- Typical inductive loads are A.C. Motors, induction furnaces, transformers and ballast-type lighting.
- Inductive loads require two kinds of power:
 - a. Active (or working) power to perform the work
 - b. Reactive power to create and maintain electro-magnetic fields

- Active power is measured in kW (Kilo Watts) and Reactive power is measured in kVAr (Kilo Volt-Amperes Reactive).



- The vector sum of the active power and reactive power make up the total (or apparent) power used.
- This is the power generated by the SEBs for the user to perform a given amount of work. Total Power is measured in kVA (Kilo Volts-Amperes).
- The vector sum of the two is called the apparent power or kVA, as illustrated above and the kVA reflects the actual electrical load on distribution system.
- The ratio of kW to kVA is called the power factor, which is always less than or equal to unity.

$$\text{Power factor } \cos \phi = \frac{\text{Active Power}}{\text{Apprarent power}} = \frac{Kw}{KVA}$$

- when electric utilities supply power, if all loads have unity power factor, maximum power can be transferred for the same distribution system capacity.
- As the loads are inductive in nature, with the power factor ranging from 0.2 to 0.9, the electrical distribution network is stressed for capacity at low power factors.

Improving Power Factor:

- The solution to improve the power factor is to add power factor correction capacitors to the plant power distribution system.
- They act as reactive power generators, and provide the needed reactive power to accomplish kW of work.
- This reduces the amount of reactive power, and thus total power, generated by the utilities.

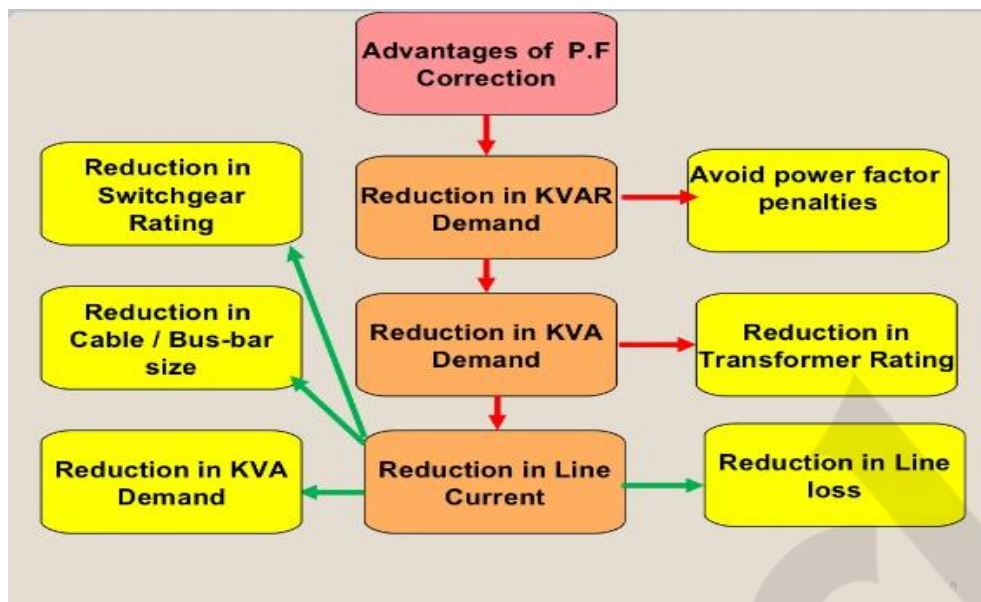
The advantages of PF improvement by capacitor addition:

- a) Reactive component of the network is reduced and so also the total current in the system from the source end.
- b) I^2R power losses are reduced in the system because of reduction in current.
- c) Voltage level at the load end is increased.
- d) kVA loading on the source generators as also on the transformers and lines upto the capacitors reduces giving capacity relief. A high power factor can help in utilising the full capacity of your electrical system.

Cost benefits of PF improvement

- a) Reduced kVA (Maximum demand) charges in utility bill.
- b) Reduced distribution losses (KWH) within the plant network.
- c) Better voltage at motor terminals and improved performance of motors.

- d) A high power factor eliminates penalty charges imposed when operating with a low power factor
- e) Investment on system facilities such as transformers, cables, switchgears etc for delivering load is reduced.



AUTOMATIC POWER FACTOR CONTROLLERS:

- Various types of automatic power factor controls are available with relay / microprocessor logic.
- Two of the most common controls are:
 1. Voltage Control and
 2. KILOVAR (kVAr) Control

1. Voltage Control:

- Voltage alone can be used as a source of intelligence when the switched capacitors are applied at point where the circuit voltage decreases as circuit load increases.
- Generally, where they are applied the voltage should decrease as circuit load increases and the drop in voltage should be around 4 – 5 % with increasing load.
- Voltage is the most common type of intelligence used in substation applications, when maintaining a particular voltage is of prime importance.
- This type of control is independent of load cycle.
- During light load time and low source voltage, this may give leading PF at the substation, which is to be taken note of.

2. KILOVAR (kVAr) Control:

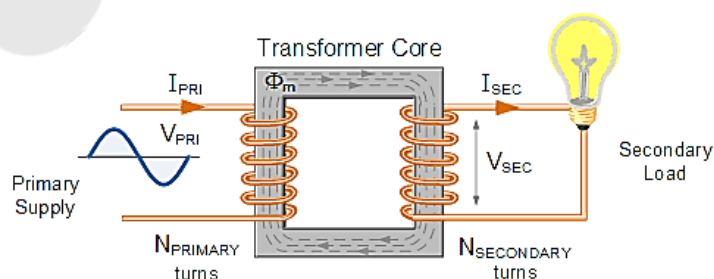
- KiloVAR sensitive controls are used at locations where the voltage level is closely regulated and not available as a control variable.
- The capacitors can be switched to respond to a decreasing power factor as a result of change in system loading.
- This type of control can also be used to avoid penalty on low power factor by adding capacitors in steps as the system power factor begins to lag behind the desired value.

- KiloVAR control requires two inputs - current and voltage from the incoming feeder, which are fed to the PF correction mechanism, either the microprocessor or the relay.

Automatic Power Factor Control Relay

- It controls the power factor of the installation by giving signals to switch on or off power factor correction capacitors.
- Relay is the brain of control circuit and needs contactors of appropriate rating for switching on/off the capacitors.
- There is a built-in power factor transducer, which measures the power factor of the installation and converts it to a DC voltage of appropriate polarity.
- This is compared with a reference voltage, which can be set by means of a knob calibrated in terms of power factor.
- When the power factor falls below setting, the capacitors are switched on in sequence.
- The relays are provided with First in First out (FIFO) and First in Last Out (FILO) sequence.
- The capacitors controlled by the relay must be of the same rating and they are switched on/off in linear sequence.
- To prevent over correction hunting, a dead band is provided.
- This setting determines the range of phase angle over which the relay does not respond; only when the PF goes beyond this range, the relay acts.
- When the load is low, the effect of the capacitors is more pronounced and may lead to hunting.
- Under current blocking (low current cut out) shuts off the relay, switching off all capacitors one by one in sequence, when load current is below setting.
- Special timing sequences ensure that capacitors are fully discharged before they are switched in. This avoids dangerous over voltage transient.
- The solid state indicating lamps (LEDS) display various functions that the operator should know and also and indicate each capacitor switching stage.

TRANSFORMER:



- A transformer can accept energy at one voltage and deliver it at another voltage.
- This permits electrical energy to be generated at relatively low voltages and transmitted at high voltages and low currents, thus reducing line losses and voltage drop.
- Transformers consist of two or more coils that are electrically insulated, but magnetically linked.
- The primary coil is connected to the power source and the secondary coil connects to the load.

- The turn's ratio is the ratio between the number of turns on the secondary to the turns on the primary.
- The secondary voltage is equal to the primary voltage times the turn's ratio.
- Ampere-turns are calculated by multiplying the current in the coil times the number of turns.
- Primary ampere-turns are equal to secondary ampere-turns.
- Voltage regulation of a transformer is the percent increase in voltage from full load to no load.

Types of Transformers:

Transformers are classified as two categories:

1. Power transformers
2. Distribution transformers.

1. Power Transformer:

Power transformers are used in transmission network of higher voltages, deployed for step-up and step down transformer application (400 kV, 200 kV, 110 kV, 66 kV, 33kV).

2. Distribution Transformer:

Distribution transformers are used for lower voltage distribution networks as a means to end user connectivity. (11kV, 6.6 kV, 3.3 kV, 440V, 230V).

Rating of Transformer:

- Rating of the transformer is calculated based on the connected load and applying the diversity factor on the connected load, applicable to the particular industry and arrive at the kVA rating of the Transformer.
- Diversity factor is defined as the ratio of overall maximum demand of the plant to the sum of individual maximum demand of various equipment.
- Diversity factor varies from industry to industry and depends on various factors such as individual loads, load factor and future expansion needs of the plant.
- Diversity factor will always be less than one.

Location of Transformer:

- Location of the transformer is very important as far as distribution loss is concerned.
- Transformer receives HT voltage from the grid and steps it down to the required voltage.
- Transformers should be placed close to the load centre, considering other features like optimization needs for centralized control, operational flexibility etc.
- This will bring down the distribution loss in cables.

Transformer Losses and Efficiency:

- The efficiency varies anywhere between 96 to 99 percent.
- The efficiency of the transformers not only depends on the design, but also, on the effective operating load.
- Transformer losses consist of two parts:

1. No-load Loss

2. Load Loss

1. No-load Loss:

- No-load loss also called core loss is the power consumed to sustain the magnetic field in the transformer's steel core.
- Core loss occurs whenever the transformer is energized.
- Core loss does not vary with load.
- Core losses are caused by two factors: hysteresis and eddy current losses.
- Hysteresis loss is that energy lost by reversing the magnetic field in the core as the magnetizing AC rises and falls and reverses direction.
- Eddy current loss is a result of induced currents circulating in the core.

2. Load Loss:

- Load loss also called copper loss is associated with full-load current flow in the transformer windings.
- Copper loss is power lost in the primary and secondary windings of a transformer due to the ohmic resistance of the windings.
- Copper loss varies with the square of the load current. $P = I^2 R$
- Therefore, total losses of the transformer is given by
- Total losses = No load losses + n^2 Full load losses

where 'n' is the load factor.

$$n = \left(\frac{\text{Load KVA}}{\text{Rated KVA}} \right)^2 \times \text{Full losses}$$

System Distribution Losses:

- In an electrical system often the constant no load losses and the variable load losses are to be assessed alongside, over long reference duration, towards energy loss estimation.
- Identifying and calculating the sum of the individual contributing loss components is a challenging one, requiring extensive experience and knowledge of all the factors impacting the operating efficiencies of each of these components.

In system distribution loss optimization, the various options available include:

- Relocating transformers and sub-stations near to load centers
- Re-routing and re-conductoring such feeders and lines where the losses / voltage drops are higher.
- Power factor improvement by incorporating capacitors at load end.
- Optimum loading of transformers in the system.
- Opting for lower resistance All Aluminum Alloy Conductors (AAAC) in place of conventional Aluminum Cored Steel Reinforced (ACSR) lines
- Minimizing losses due to weak links in distribution network such as jumpers, loose contacts, old brittle conductors.

Energy Efficient Transformers:

- Most energy loss in dry-type transformers occurs through heat or vibration from the core.
- The new high-efficiency transformers minimise these losses.
- The conventional transformer is made up of a silicon alloyed iron core.
- The iron loss of any transformer depends on the type of core used in the transformer.
- However, the latest technology is to use amorphous material - a metallic glass alloy for the core.
- The expected reduction in energy loss over conventional transformers is roughly around 70%, which is quite significant.
- By using an amorphous core with unique physical and magnetic properties, these new type of transformers have increased efficiencies even at low loads – 98.5% efficiency at 35% load.
- Electrical distribution transformers made with amorphous metal cores provide excellent opportunity to conserve energy right from the installation.
- Though these transformers are a little costlier than conventional iron core transformers, the overall benefit towards energy savings will compensate for the higher initial investment. At present amorphous metal core transformers are available up to 1600 kVA.

ENERGY EFFICIENT MOTORS:

Minimizing Watts Loss in Motors:

- Improvements in motor efficiency can be achieved without compromising motor performance - at higher cost - within the limits of existing design and manufacturing technology.
- The improvement in motor efficiency must result from reducing the Watts losses.
- In terms of the existing state of electric motor technology, a reduction in watts losses can be achieved in various ways.
- All of these changes to reduce motor losses are possible with existing motor design and manufacturing technology.
- They would, however, require additional materials and/or the use of higher quality materials and improved manufacturing processes resulting in increased motor cost.

Reduced Losses = Improved Efficiency

Watts Loss Area	Efficiency Improvement
1. Iron	Use of thinner gauge, lower loss core steel reduces eddy current losses. Longer core adds more steel to the design, which reduces losses due to lower operating flux densities.
2. Stator $I^2 R$	Use of more copper and larger conductors increases cross sectional area of stator windings. This lowers resistance (R) of the windings and reduces losses due to current flow (I).
3. Rotor $I^2 R$	Use of larger rotor conductor bars increases size of cross section, lowering conductor resistance (R) and losses due to current flow (I).
4. Friction & Windage	Use of low loss fan design reduces losses due to air movement.
5. Stray Load Loss	Use of optimised design and strict quality control procedures minimizes stray load losses.

- Thus energy-efficient electric motors reduce energy losses through improved design, better materials, and improved manufacturing techniques.
- Replacing a motor may be justifiable solely on the electricity cost savings derived from an energy-efficient replacement.
- This is true if the motor runs continuously, power rates are high, the motor is oversized for the application, or its nominal efficiency has been reduced by damage or previous rewinds.

Technical aspects of Energy Efficient Motors:

- Energy-efficient motors last longer, and may require less maintenance. At lower temperatures, bearing grease lasts longer; required time between re-greasing increases. Lower temperatures translate to long lasting insulation. Generally, motor life doubles for each 10°C reduction in operating temperature.
- Select energy-efficient motors with a 1.15 service factor, and design for operation at 85% of the rated motor load.
- Electrical power problems, especially poor incoming power quality can affect the operation of energy-efficient motors.
- Speed control is crucial in some applications. In polyphase induction motors, slip is a measure of motor winding losses. The lower the slip, the higher the efficiency. Less slippage in energy efficient motors results in speeds about 1% faster than in standard counterparts.
- Starting torque for efficient motors may be lower than for standard motors. Facility managers should be careful when applying efficient motors to high torque applications.

SOFT STARTER:

- When starting, AC Induction motor develops more torque than is required at full speed.
- This stress is transferred to the mechanical transmission system resulting in excessive wear and premature failure of chains, belts, gears, mechanical seals, etc.
- Additionally, rapid acceleration also has a massive impact on electricity supply charges with high inrush currents drawing +600% of the normal run current.
- The use of Star Delta only provides a partial solution to the problem. Should the motor slow down during the transition period, the high peaks can be repeated and can even exceed direct on line current.
- Soft starter provides a reliable and economical solution to these problems by delivering a controlled release of power to the motor, thereby providing smooth, step less acceleration and deceleration.
- Motor life will be extended as damage to windings and bearings is reduced.
- Soft Start & Soft Stop is built into 3 phase units, providing controlled starting and stopping with a selection of ramp times and current limit settings to suit all applications

Advantages of Soft Start

- Less mechanical stress

- Improved power factor
- Lower maximum demand
- Less mechanical maintenance

VARIABLE SPEED DRIVES:

Speed Control of Induction Motors

- Induction motor is the workhorse of the industry.
- It is cheap rugged and provides high power to weight ratio.
- On account of high cost-implications and limitations of D.C. System, induction motors are preferred for variable speed application, the speed of which can be varied by changing the supply frequency.
- The speed can also be varied by varying the input voltage, varying the resistance of the rotor circuit, using multi speed windings, using Scherbius or Kramer drives, using mechanical means such as gears and pulleys and eddy-current or fluid coupling, or by using rotary or static voltage and frequency converters.

Variable Frequency Drive

- The VFD operates on a simple principle. The rotational speed of an AC induction motor depends on the number of poles in that stator and the frequency of the applied AC power.
- Although the number of poles in an induction motor cannot be altered easily, variable speed can be achieved through a variation in frequency.
- The VFD rectifies standard 50 cycle AC line power to DC, then synthesizes the DC to a variable frequency AC output.
- Motors connected to VFD provide variable speed mechanical output with high efficiency.
- These devices are capable of up to a 9:1 speed reduction ratio (11 percent of full speed), and a 3:1 speed increase (300 percent of full speed).
- The technology of AC variable frequency drives (VFD) has evolved into highly sophisticated digital microprocessor control, along with high switching frequency IGBTs (Insulated Gate Bi Polar Transistors) power devices.
- This has led to significantly advanced capabilities from the ease of programmability to expanded diagnostics.
- The two most significant benefits from the evolution in technology have been that of cost and reliability, in addition to the significant reduction in physical size.

Variable Torque Vs. Constant Torque

- Variable speed drives, and the loads that are applied to, can generally be divided into two groups:
 1. Constant torque
 2. Variable torque.

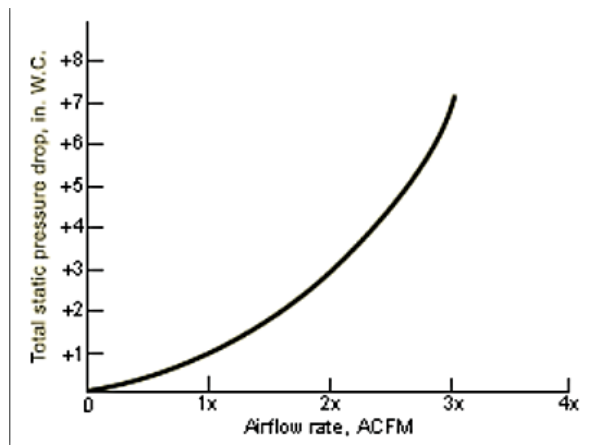
- The energy savings potential of variable torque applications is much greater than that of constant torque applications.
- Constant torque loads include vibrating conveyors, punch presses, rock crushers, machine tools, and other applications where the drive follows a constant V/Hz ratio.
- Variable torque loads include centrifugal pumps and fans, which make up the majority of HVAC applications.

PERFORMANCE EVALUATION OF FANS:

- Fans and blowers provide air for ventilation and industrial process requirements.
- Fans generate a pressure to move air (or gases) against a resistance caused by ducts, dampers, or other components in a fan system.
- The fan rotor receives energy from a rotating shaft and transmits it to the air.

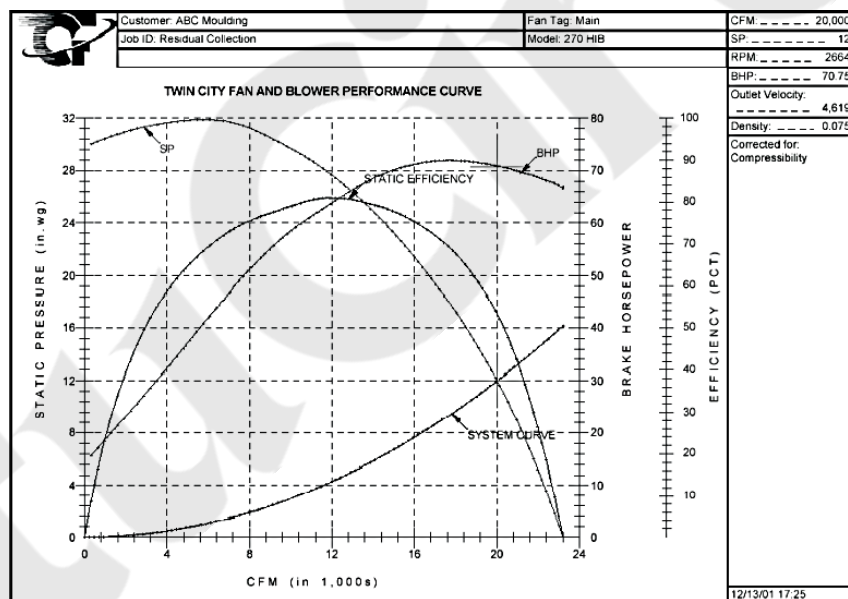
System Characteristics:

- The term "system resistance" is used when referring to the static pressure.
- The system resistance is the sum of static pressure losses in the system.
- The system resistance is a function of the configuration of ducts, pickups, elbows and the pressure drops across equipment for example backfilter or cyclone.
- The system resistance varies with the square of the volume of air flowing through the system.
- For a given volume of air, the fan in a system with narrow ducts and multiple short radius elbows is going to have to work harder to overcome a greater system resistance than it would in a system with larger ducts and a minimum number of long radius turns.
- Long narrow ducts with many bends and twists will require more energy to pull the air through them.
- For a given fan speed, the fan will be able to pull less air through this system than through a short system with no elbows.
- Thus, the system resistance increases substantially as the volume of air flowing through the system increases; square of air flow.
- Conversely, resistance decreases as flow decreases. To determine what volume the fan will produce, it is therefore necessary to know the system resistance characteristics.
- Typically a system resistance curve is generated with for various flow rates on the x-axis and the associated resistance on the y-axis.



Fan Characteristics:

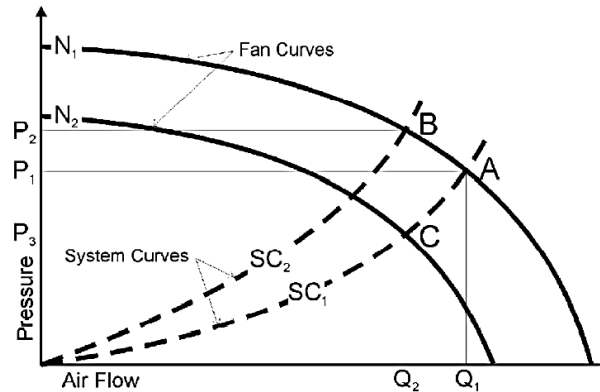
- Fan characteristics can be represented in form of fan curve(s).
- The fan curve is a performance curve for the particular fan under a specific set of conditions.
- The fan curve is a graphical representation of a number of inter-related parameters.
- Typically, a curve will be developed for a given set of conditions usually including: fan volume, system static pressure, fan speed, and brake horsepower required to drive the fan under the stated conditions.



- Some fan curves will also include an efficiency curve so that a system designer will know where on that curve the fan will be operating under the chosen conditions.
- The curve static pressure (SP) vs. flow is especially important.
- The intersection of the system curve and the static pressure curve defines the operating point.
- When the system resistance changes, the operating point also changes.
- Once the operating point is fixed, the power required could be found by following a vertical line that passes through the operating point to an intersection with the power (BHP) curve.
- A horizontal line drawn through the intersection with the power curve will lead to the required power on the right vertical axis.
- In the depicted curves, the fan efficiency curve is also presented.

System Characteristics and Fan Curves:

- In any fan system, the resistance to air flow (pressure) increases when the flow of air is increased.
- It varies as the square of the flow.
- The pressure required by a system over a range of flows can be determined and a "system performance curve" can be developed.



- This system curve can then be plotted on the fan curve to show the fan's actual operating point at "A" where the two curves (N₁ and SC₁) intersect.
- This operating point is at air flow Q₁ delivered against pressure P₁.
- A fan operates along a performance given by the manufacturer for a particular fan speed.
- At fan speed N₁, the fan will operate along the N₁ performance curve as shown in Figure.
- The fan's actual operating point on this curve will depend on the system resistance; fan's operating point at "A" is flow (Q₁) against pressure (P₁).
- Two methods can be used to reduce air flow from Q₁ to Q₂:

First method:

- First Method is to restrict the air flow by partially closing a damper in the system.
- This action causes a new system performance curve (SC₂) where the required pressure is greater for any given air flow.
- The fan will now operate at "B" to provide the reduced air flow Q₂ against higher pressure P₂.

Second method:

- Second method to reduce air flow is by reducing the speed from N₁ to N₂, keeping the damper fully open.
- The fan would operate at "C" to provide the same Q₂ air flow, but at a lower pressure P₃.
- Thus, reducing the fan speed is a much more efficient method to decrease airflow since less power is required and less energy is consumed.

Fan Performance Assessment:

The fans are tested for field performance by measurement of flow, head, temperature on the fan side and electrical motor kW input on the motor side.

➤ Air flow measurement

- **Static pressure:**

Static pressure is the potential energy put into the system by the fan. It is given up to friction in the ducts and at the duct inlet as it is converted to velocity pressure. At the inlet to the duct, the static pressure produces an area of low pressure

- **Velocity pressure:**

Velocity pressure is the pressure along the line of the flow that results from the air flowing through the duct. The velocity pressure is used to calculate air velocity.

- **Total pressure:**

Total pressure is the sum of the static and velocity pressure. Velocity pressure and static pressure can change as the air flows through different size ducts, accelerating and decelerating the velocity. The total pressure stays constant, changing only with friction losses.

- **Measurement by Pitot tube:**

Total pressure is measured using the inner tube of pitot tube and static pressure is measured using the outer tube of pitot tube. When the inner and outer tube ends are connected to a manometer, we get the velocity pressure. For measuring low velocities, it is preferable to use an inclined tube manometer instead of U tube manometer.

FLOW CONTROL STRATEGIES:

- Once a fan system is designed and installed, the fan operates at a constant speed.
- There may be occasions when a speed change is desirable, i.e., when adding a new run of duct that requires an increase in air flow (volume) through the fan. There are also instances when the fan is oversized and flow reductions are required.
- Various ways to achieve change in flow are: pulley change, damper control, inlet guide vane control, variable speed drive and series and parallel operation of fans.

Pulley Change

- When a fan volume change is required on a permanent basis, and the existing fan can handle the change in capacity, the volume change can be achieved with a speed change.
- The simplest way to change the speed is with a pulley change.
- For this, the fan must be driven by a motor through a v-belt system.
- The fan speed can be increased or decreased with a change in the drive pulley or the driven pulley or in some cases, both pulleys.

Damper Controls

- Some fans are designed with damper controls. Dampers can be located at inlet or outlet.
- Dampers provide a means of changing air volume by adding or removing system resistance.
- This resistance forces the fan to move up or down along its characteristic curve, generating more or less air without changing fan speed.
- However, dampers provide a limited amount of adjustment, and they are not particularly energy efficient.

Inlet Guide Vanes

- Inlet guide vanes are another mechanism that can be used to meet variable air demand.
- Guide vanes are curved sections that lay against the inlet of the fan when they are open.
- When they are closed, they extend out into the air stream.
- As they are closed, guide vanes pre-swirl the air entering the fan housing.
- This changes the angle at which the air is presented to the fan blades, which, in turn, changes the characteristics of the fan curve.
- Guide vanes are energy efficient for modest flow reductions – from 100 percent flow to about 80 percent. Below 80 percent flow, energy efficiency drops sharply.
- Axial-flow fans can be equipped with variable pitch blades, which can be hydraulically or pneumatically controlled to change blade pitch, while the fan is at stationary.
- Variable-pitch blades modify the fan characteristics substantially and thereby provide dramatically higher energy efficiency than the other options discussed thus far.

Variable Speed Drives

- Variable speed drives are expensive, they provide almost infinite variability in speed control.
- Variable speed operation involves reducing the speed of the fan to meet reduced flow requirements.
- Fan performance can be predicted at different speeds using the fan laws. Since power input to the fan changes as the cube of the flow, this will usually be the most efficient form of capacity control.
- However, variable speed control may not be economical for systems, which have infrequent flow variations.

Series and Parallel Operation

- Parallel operation of fans is another useful form of capacity control.
- Fans in parallel can be additionally equipped with dampers, variable inlet vanes, variable-pitch blades, or speed controls to provide a high degree of flexibility and reliability.
- Combining fans in series or parallel can achieve the desired airflow without greatly increasing the system package size or fan diameter.
- Parallel operation is defined as having two or more fans blowing together side by side. The performance of two fans in parallel will result in doubling the volume flow, but only at free delivery.
- Series operation can be defined as using multiple fans in a push-pull arrangement. By staging two fans in series, the static pressure capability at a given airflow can be increased.
- In both series and parallel operation, particularly with multiple fans certain areas of the combined performance curve will be unstable and should be avoided.

ENERGY SAVING/ CONSERVATION OPPORTUNITIES IN FAN

1. Minimising excess air level in combustion systems to reduce FD fan and ID fan load.
2. Minimising air in-leaks in hot flue gas path to reduce ID fan load, especially in case of kilns, boiler plants, furnaces, etc. Cold air in-leaks increase ID fan load tremendously, due to density increase of flue gases and in-fact choke up the capacity of fan, resulting as a bottleneck for boiler / furnace itself.
3. In-leaks / out-leaks in air conditioning systems also have a major impact on energy efficiency and fan power consumption and need to be minimized.
4. The findings of performance assessment trials will automatically indicate potential areas for improvement, which could be one or a more of the following:
 - Change of impeller by a high efficiency impeller along with cone.
 - Change of fan assembly as a whole, by a higher efficiency fan
 - Impeller de-rating (by a smaller dia impeller)
 - Change of metallic / Glass reinforced Plastic (GRP) impeller by the more energy efficient hollow FRP impeller with aerofoil design, in case of axial flow fans, where significant savings have been reported.
 - Fan speed reduction by pulley dia modifications for derating.
 - Option of two speed motors or variable speed drives for variable duty conditions.
 - Option of energy efficient flat belts, or, cogged raw edged V belts, in place of conventional V belt systems, for reducing transmission losses.
 - Adopting inlet guide vanes in place of discharge damper control.
 - Minimizing system resistance and pressure drops by improvements in duct system.

ELECTRONIC BALLAST:

Role of Ballast

- In an electric circuit the ballast acts as a stabilizer.
- Fluorescent lamp is an electric discharge lamp.
- The two electrodes are separated inside a tube with no apparent connection between them.
- When sufficient voltage is impressed on these electrodes, electrons are driven from one electrode and attracted to the other.
- The current flow takes place through an atmosphere of low pressure mercury vapour.
- Since the fluorescent lamps cannot produce light by direct connection to the power source, they need an ancillary circuit and device to get started and remain illuminated.
- The auxillary circuit housed in a casing is known as ballast

Conventional Vs Electronic Ballasts

- The conventional ballasts make use of the kick caused by sudden physical disruption of current in an inductive circuit to produce the high voltage required for starting the lamp and then rely on reactive voltage drop in the ballast to reduce the voltage applied across the lamp.

- On account of the mechanical switch (starter) and low resistance of filament when cold the uncontrolled filament current, generally tend to go beyond the limits specified by Indian standard specifications.
- With high values of current and flux densities the operational losses and temperature rise are on the higher side in conventional choke.

The basic functions of electronic ballast are:

- To ignite the lamp
- To stabilize the gas discharge
- To supply the power to the lamp

ENERGY EFFICIENT LIGHTING CONTROLS:

Occupancy Sensors

- Occupancy-linked control can be achieved using infra-red, acoustic, ultrasonic or microwave sensors, which detect either movement or noise in room spaces.
- These sensors switch lighting on when occupancy is detected, and off again after a set time period, when no occupancy movement detected.
- They are designed to override manual switches and to prevent a situation where lighting is left on in unoccupied spaces.

Timed Based Control

- Timed-turnoff switches are the least expensive type of automatic lighting control.
- In some cases, their low cost and ease of installation makes it desirable to use them where more efficient controls would be too expensive.

Types and Features:

- The oldest and most common type of timed-turnoff switch is the "dial timer," a spring-wound mechanical timer that is set by twisting the knob to the desired time.
- Typical units of this type are vulnerable to damage because the shaft is weak and the knob is not securely attached to the shaft.
- Some spring-wound units make an annoying ticking sound as they operate.
- Newer types of timed-turnoff switches are completely electronic and silent.
- Electronic switches can be made much more rugged than the spring-wound dial timer.
- These units typically have a spring-loaded toggle switch that turns on the circuit for a preset time interval.
- Some electronic models provide a choice of time intervals, which you select by adjusting a knob located behind the faceplate

- Most models allow occupants to turn off the lights manually. Some models allow occupants to keep the lights on, overriding the timer
- Timed-turnoff switches are available with a wide range of time spans.
- The choice of time span is a compromise. Shorter time spans waste less energy but increase the probability that the lights will turn off while someone is in the space.
- Dial timers allow the occupant to set the time span, but this is not likely to be done with a view toward optimizing efficiency.
- For most applications, the best choice is an electronic unit that allows the engineering staff to set a fixed time interval behind the cover plate.

Daylight Linked Control:

- Photoelectric cells can be used either simply to switch lighting on and off, or for dimming.
- They may be mounted either externally or internally.
- It is however important to incorporate time delays into the control system to avoid repeated rapid switching caused, for example, by fast moving clouds.
- By using an internally mounted photoelectric dimming control system, it is possible to ensure that the sum of daylight & electric lighting always reaches the design level by sensing the total light in the controlled area and adjusting the output of the electric lighting accordingly.
- If daylight alone is able to meet the design requirements, then the electric lighting can be turned off.
- The energy saving potential of dimming control is greater than a simple photoelectric switching system.
- Dimming control is also more likely to be acceptable to room occupants.

Localized Switching

- Localized switching should be used in applications which contain large spaces.
- Local switches give individual occupants control over their visual environment and also facilitate energy savings.
- By using localized switching it is possible to turn off artificial lighting in specific areas, while still operating it in other areas where it is required, a situation which is impossible if the lighting for an entire space is controlled from a single switch.